

# **Data Analyst Course Details**

## **100% Practical Classes**

### **Session 1**

**Installation of Sql Server management studio**

### **Session 2**

**Introduction To SQL Server**

- Connecting To Server
- Sql Server Management Studio and Tools In Management Studio
- Object Explorer
- Database
- Constraints

### **Session 3**

**Introduction To TSQL**

- History and Features of TSQL
- Types Of TSQL Commands
- Data Definition Language (DDL)
- Creating Database
- Altering Database
- Deleting Database
- Primary Key and Referential Integrity or foreign key constraints
- Data Types In TSQL
- Table

- Creating Table
- Altering Table
- Deleting Table

## Session 4

- Data Manipulation Language (DML)
- Data Query Language (DQL)

### **Data Manipulation Language**

- Insert
- Identity
- Update
- Delete
- Truncate
- Differences Between Delete and Truncate

## Session 5

### **Data Query Language (DQL)**

- Select
- Where clause
- Order By Clause
- Distinct Keyword
- Column aliases
- Between ... And
- In
- Like
- Is Null
- **Sub queries**

## Session 6

### Built In Functions

- Aggregate Functions
- Group By and Having Clauses

## Session 7

### SQL Operators

#### Set Operators

- Union
- Union All
- Intersect
- Except

## Session 8

### Joins

- Inner Join
- Outer Join
- Left Outer Join
- Right Outer Join
- Full Outer Join
- Cross Join

## Session 9 & 10

### Test / Assignments

# SSIS

**100 % Practical Classes**

## Session 11

### **Introduction to SQL Server Integration Services**

- SSIS Package Architecture Overview
- Introduction to SSIS & Data Warehouse
- Basic ETL Entities in SSIS
- Data and Management Tools
- Installation of the tool

## Session 12

### **Introduction to Data Flow**

- Data Flow Overview
- Data Sources
- Data Destinations
- Data Flow Transformations
- Data Viewers

### **Data Sources**

- Excel Source
- Flat File Source
- OLE DB Source
- XML Source

### **Data Flow Transformations**

- Conditional Split Transformation

## Session 13

- Data Conversion Transformation
- Aggregate Transformation

## Session 14

- Derived Column Transformation
- Lookup Transformation

## Session 15

- Sort Transformation
- Merge Transformation
- Union All Transformation

## Session 16

- Multicast Transformation
- Merge Join Transformation
- Slowly Changing Dimension Transformation

## Session 17

- Audit Transformation
- Character Map Transformation
- Copy Column Transformation
- Import Transformation

## Session 18

- Export Transformation
- Fuzzy Lookup Transformation
- Fuzzy Grouping Transformation

### Data Flow Destinations

- Data Reader Destination
- Excel Destination
- Flat File Destination
- OLE DB Destination

## Session 19

### Introduction to Control Flow

- Control Flow Overview

- The Execute SQL Task
- The File System Task
- Backup database Task
- Data Flow Task

## **Session 20**

- Execute Package Task
- Execute Process Task

## **Session 21**

### **Error Handling and Logging**

- Control Flow: The On Error event handler
- Data Flow: Error data flow
- Built-in log providers

### **SSIS Package Management**

- The SSIS Service
- Managing packages with SQL Server Management Studio
- Scheduling packages with SQL Server Agent

## **Session 22 & 23**

### **Projects / Assignments**

## **SSAS**

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## **Session 24**

**What Is Microsoft Business Intelligence?**

- Defining Microsoft Business Intelligence
- Viewing a Cube Using Excel
- Using SSAS in SSDT
- Understanding SSDT
- Creating Data Sources
- Creating Data Source Views
- Creating a multidimensional cube

## Session 25

### Intermediate SSAS

- KPI
- Action
- Aggregation
- Creating Perspectives
- Creating Translations
- Working with Multiple Fact Tables
- Using the Business Intelligence Wizard
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## Session 26

- Creating a TabularModel
- KPI
- Aggregation

## SSRS

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## Session 27

### Creating basic reports

- Tabular
- List
- Matrix
- Chart
- Parameterized
- Drilldown

- DrillThrough
- Constructing data sources and Datasets

## **Power BI**

### **100 % Practical Classes**

## **Session 28**

### **Module : Power BI Desktop & Power Query**

- Getting Started
- Class Introduction (BI Basics)
- Class Introduction (Power BI Desktop Overview)
- Data Discovery with Power BI Desktop
- Power View
- Power Pivot
- Power Query
- Pie Chart
- Donut Chart
- Treemap Chart
- Table Chart

## **Session 29**

- Types of Filters
- Drill Down Report
- Drill Through Report

## **Session 30**

### **Module : Get Data**



- Excel
- CSV
- Sql Server

## Session 31

- Creating Calculated Columns and Tables (Navigation Function)
- Creating Calculated Columns and Tables (Calculated Tables)
- Creating Calculated Measures (Measure Basics)
- Manual refresh
- Text Box
- More Visuals
- Publish

## Session 32

- Insert Tab
- Introduction to Modeling Data
- Creating the Data Model (Modeling Basics)

## Session 33

- View Tab
- Mobile BI (Power BI Mobile Overview)
- Mobile BI (Designing Reports and Dashboards for Mobile)
- Mobile BI (Interacting with the Power BI Mobile App)
- Format Tab

## Session 34

- Data Getaway
- Refreshing the Data (Refreshing Data Overview)

- Refreshing the Data (Installing the Data Gateway)
- Refreshing the Data (Scheduling a Data Refresh)

## **Session 35**

- Power Query
- Transforming Data (Add Column From Example)
- Transforming Data (Appending Queries)
- Transforming Data (Merging Queries)
- Transforming Data (Combine Files)
- Transforming Data (Other Query Features)

## **Session 36 & 37**

### **Projects / Assignments**